

Opening Offers and Bargaining Power in Two LLM Markets

Q Competitive Market Behavior of LLMs [arXiv:2609.02580](#)

P Commitment To Cooperation With Self-Negotiated Contracts [arXiv:2607.22750](#)

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract

Paper Q studies language-model traders in a double auction, while paper P studies their negotiated contracts in a resource game. This thread asked whether P's apparent advantage for one player reflects a better option outside the bargain, the chance to propose first, or both. Q reports that later quotes respond to an opening offer, and P reports favourable point contracts for the player who always proposes first, but the papers do not separate the two causes in P. The thread paused because that separation needs a small, repeated experiment in which player role and proposal order vary independently.

1 Introduction

Q places language-model buyers and sellers in a continuous double auction. A buyer values a unit, a seller bears a cost, and a trade occurs as soon as a bid and an ask cross. Q reports slow or absent convergence towards market equilibrium and fewer trades than the market could support. It also reports that quotes from agents on the other side track the first quote, even though the trading rule does not require those agents to follow it [5].

P places two language-model players, P-Red and P-Blue, on a board where they need coloured chips to reach a goal. They can trade chips and negotiate contracts that enforce later transfers of chips or finishing points. Contracts can improve cooperation when a player must first help another player and wait for a return. On P's asymmetric boards, P-Red can reach the goal alone, while P-Blue needs help. P-Red also opens every contract negotiation [7].

The connection pursued was a question about bargaining power. An independent player can refuse a poor deal and still finish; a first proposer can also set the starting terms. Q's opening-quote pattern suggested that proposal order might matter in P even when neither player has a stronger option outside the contract. That suggestion is a reason to test, not evidence that Q's market and P's game share a causal mechanism.

2 What was done

The peers first asked whether P's enforceable contracts could repair Q's missed trades. They checked Q's transaction rule against P's reason for using contracts. In Q, a crossing quote executes at once. In P, a promise to help later may be broken unless a contract enforces it. The peers therefore set aside a direct transfer of P's commitment remedy to Q. Q instead connects low trade volume to small successive quote changes and agents' reluctance to cross before the round ends [5, 7].

One peer then considered whether contract-like wording could change behaviour even when Q's offers already bind. Another proposed comparing total gains with each trader's gain over an outside option. Checks against Q's prompts and P's measure of mutual benefit narrowed both ideas. The thread finally focused on the cleaner question: in P's point contracts, how much of P-Red's advantage comes from being able to finish independently, and how much from proposing first? The peers checked Q's opening-quote result, P's fixed assignment of roles and proposer, and P's own discussion of first-mover advantage [5, 7].

The peers also made limited prior-work searches. The ledger records studies of anchoring in language-model negotiation [6, 4] and adjacent work on binding offers, natural-language contracts, and supply-chain bargaining [1, 3, 2]. Those searches did not establish that the proposed experiment is novel.

3 Results

The transaction rules give a narrow result about the first idea. Let v be a buyer's value for one unit, c the seller's cost, and p the price of an immediate trade. Their payoffs are $(v - p, p - c)$. If a contract merely transfers t from the buyer to the seller when that same trade occurs, the payoffs become

$$(v - p - t, p + t - c).$$

These are exactly the payoffs from an immediate trade at the effective price $p + t$. Under the peers' stated assumptions of one unit, feasible prices, and no later or third-party obligations, the transfer adds no new payoff possibility. P's contracts solve a different problem: they make later performance credible. This argument does not rule out a change in trading behaviour caused by the wording of an offer [5, 7].

For the main question, P reports that P-Red receives about 60% of negotiated points in its programmatic points contracts even on independent and mutually dependent boards, where P-Red lacks the asymmetric outside-option advantage. P calls this a possible first-mover advantage. On asymmetric boards, however, P-Red is both the independent player and the first proposer. Its favourable contract terms there cannot be divided into a role effect and an order effect from the reported comparisons. P's finding that full state visibility improves P-Red's terms on asymmetric boards leaves the same distinction unresolved [7]. Q's response to opening quotes makes an order effect plausible, but it supplies no estimate of that effect in P [5].

4 What did not hold

The peers considered an objection based on individual losses hidden by aggregate gains. Q's exchange permits loss-making offers, but Q's prompts tell buyers not to bid above their value and sellers not to ask below their cost. If agents obey those instructions, an executed trade gives each side a non-negative gain. Q does not report how often agents disobey them, so

the supplied material does not show a widespread loss problem [5]. P’s measure requiring both players to beat their baseline is strict: it also fails when one player merely matches that baseline. A zero on that measure alone does not prove that either player was harmed [7].

The other proposed link also remained open. P reports less subsequent trading with natural-language tile contracts than with programmatic tile contracts, despite similar numbers of covered moves. Q reports opening-quote responses in a market with immediate settlement [7, 5]. These are observations in different games. They do not show that contract language would reduce trades in Q, or that cognitive anchoring caused P’s point-contract split. Likewise, an early proposal and a favourable final deal can move together without proving why the agents chose those terms.

5 Why the thread stopped

The verifier paused the thread because the main connection is still a hypothesis. Q’s opening-quote pattern does not establish an effect of proposal order on P’s contract bargains, and P already notes the possible first-mover advantage. The supplied work contains no outcomes from a test that separates order from the independent player’s outside option.

The smallest restart is to run P’s programmatic points contract on matched original and red/blue-swapped asymmetric boards, while independently randomising which player proposes first and repeating matched runs. Keep the model pairing, board geometry, negotiation limit, and judge fixed. Compare net point transfers and final rewards to the independent player across the four role-by-order combinations; also record agreement rates, joint reward, gains over baseline, and opening proposals. An independent-player premium when that player proposes second would support an outside-option effect. A proposer premium within either role would support an order effect. Opening proposals would help describe the path to agreement, but would not by themselves prove cognitive anchoring.

References

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